Virasoro TQFT as a description of pure 3D gravity

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August 21, 2025

Outline

1. Introduction
2. Overview
3. 3d gravity as a gauge theory
4. Phase space and the Teichmüller component
5. Quantizing Teichmüller space
6. Mapping class group action
7. 3d gravity and Virasoro TQFT

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- I will review a recent proposed solution to this problem when $\Lambda < 0$ (Collier et al., 2023).

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- The gravity phase space is equal to two copies of the Teichmüller space $\mathcal{T}$ of the spatial slice Σ (Krasnov and Schlenker, 2007; Scarinci and Krasnov, 2013), i.e.,

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- Infinite-dimensional Hilbert space of *normalizable* Virasoro conformal blocks on Σ.
- The mapping class group of Σ, i.e., modular transformations, has a unitary action on this Hilbert space, implementing crossing transformations.
- Defines a 3D TQFT called Virasoro TQFT (Ellegaard Andersen and Kashaev, 2014).

Note. The TQFT we consider in this paper has sometimes been referred to as "Teichmüller TQFT" in the literature. We prefer the name "Virasoro TQFT" because we feel it better captures the physical meaning of the TQFT, and we strongly feel that a person like O. Teichmüller does not deserve additional scientific honor for a recent development that he did not contribute to. Moreover our treatment of

Figure

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- Generalization of the Maloney-Witten sum (Maloney and Witten, 2010).

3d gravity as a gauge theory

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$$\int \left\{ e^a \wedge \left(d\omega_a + \frac{1}{2}\epsilon_{abc}\,\omega^b \wedge \omega^c \right) - \frac{\Lambda}{6}\epsilon_{abc}\,e^a \wedge e^b \wedge e^c \right\}, \tag{4}$$

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where e^a is the dreibein and

$$\omega_a = \frac{1}{2}\epsilon_{abc}\omega^{bc}. \tag{5}$$

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We can introduce gauge fields

$$A_a^{\pm} = \omega_a \pm \frac{1}{l} e_a, \quad \text{with} \quad \Lambda = -\frac{1}{l^2}. \tag{6}$$

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where

$$I_{CS}[A] = \frac{k}{4\pi} \int \mathrm{Tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right) \tag{8}$$

is the Chern-Simons action with level $k = l/4G$.

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Upshot

Virasoro TQFT solves the second problem. The other problems are *fixed by hand*.

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- The phase space of 3d gravity is two copies of Teichmüller space:

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where H is the mean curvature.

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- Teichmüller space is

$$\mathcal{T} = \mathcal{E}_{cc}/(LL \times \text{Diff}_0), \tag{14}$$

where $\mathcal{E}_{cc}$ is constant curvature zweibeins, i.e., $d\omega = \Lambda\, e^+ \otimes e^-$.

Quantizing Teichmüller space

- On $\mathcal{E}_{cc}$, the symplectic form

$$\Omega = \frac{k}{2\pi} \int_{\Sigma} \left[\delta\omega(e^+, e^-) \wedge \delta\omega(e^+, e^-) - \Lambda\, \delta e^+ \wedge \delta e^- \right] \quad (15)$$

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  where

$$Z_{\mathsf{TLL}} = \int[d\phi]\exp\left[\frac{i}{48\pi}(c-26)\int_\Sigma\left(\partial\phi\,\bar\partial\phi + \Lambda e^{\phi}\right)\right]. \quad (17)$$

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Pair of pants decomposition

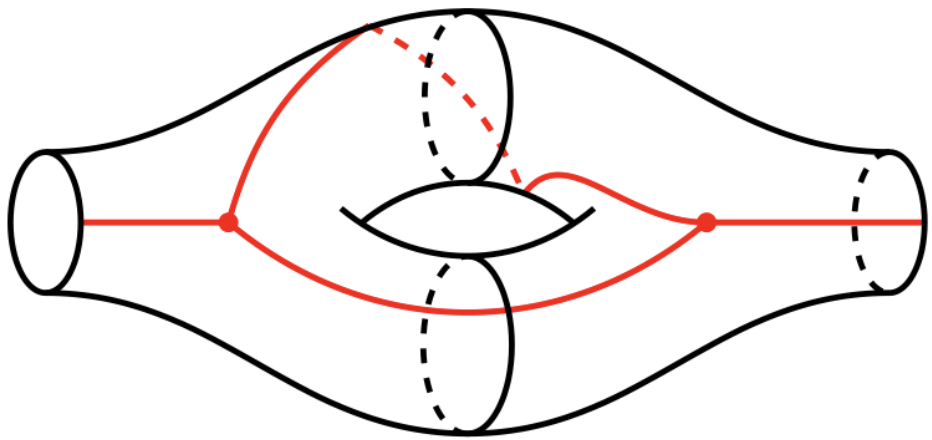


Figure: Pair of pants decomposition with a red dual graph.

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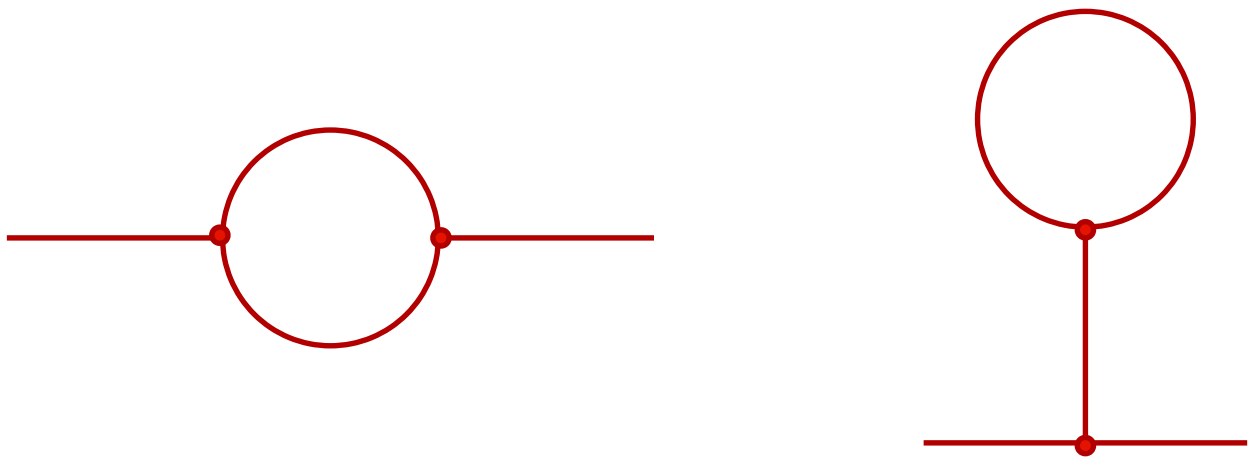


Figure: Genus 1 surface with two punctures.

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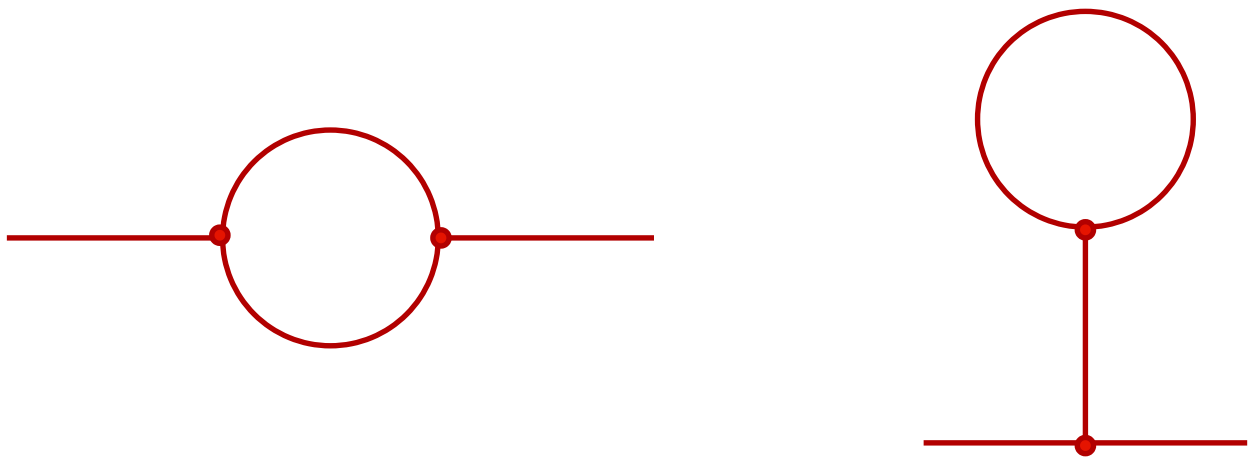


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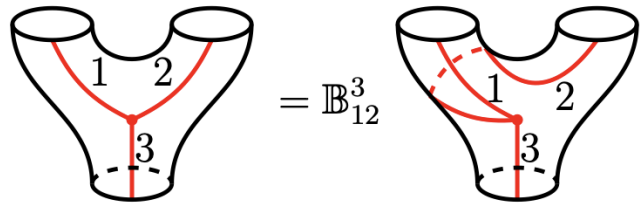
Requirement

The mapping class group must have a unitary action on the Virasoro conformal blocks.

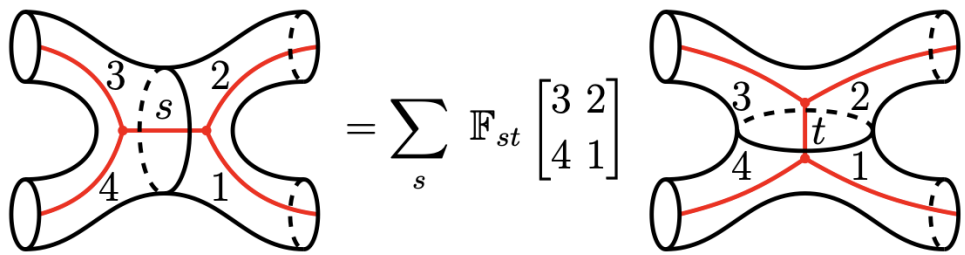
Moore-Seiberg groupoid

The following moves take us from one channel to another.

1. Braiding:



2. Fusion:



3. Torus modular S-transform:

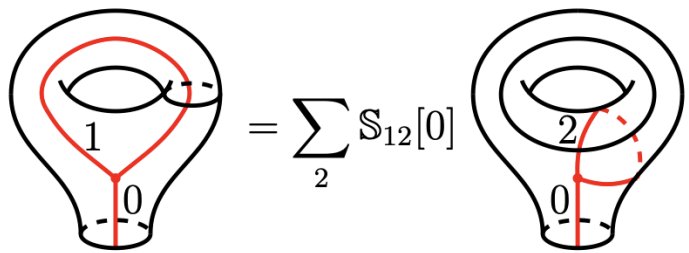


Figure: Basic moves to go between channels (Moore and Seiberg, 1989).

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Virasoro TQFT

This data allows us to define a 3d TQFT called Virasoro TQFT, similar to Chern-Simons/WZW correspondence (Elitzur et al., 1989).

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Not always well-defined!

In all our discussion, we tacitly assumed $g \geq 2$. Sphere with less than 3 punctures and torus with no punctures are not defined.

3d gravity and Virasoro TQFT

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Euclidean gravity path integral

Want to compute

$$\sum_{\text{topologies}} \int \frac{[dg]}{\text{Diff}(M, \partial M)} e^{-S[g]}$$

with fixed boundary conditions.

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- In Virasoro TQFT, we only mod out by Diff$_0(M, \partial M)$.
- The difference is captured by

$$\text{Map}(M, \partial M) = \text{Diff}(M, \partial M)/\text{Diff}_0(M, \partial M). \tag{19}$$

3d gravity and Virasoro TQFT

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Further simplification

For any $\gamma \in \mathrm{Map}(M, \partial M) \subset \mathrm{Map}(\partial M)$, we have

$$Z_{\mathrm{Vir}}(M) = Z_{\mathrm{Vir}}(M^\gamma). \tag{21}$$

3d gravity and Virasoro TQFT

Therefore

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where

$$\mathsf{Map}_0(M,\partial M) = \frac{\mathsf{Map}(M,\partial M)}{\ker(\mathsf{Map}(M,\partial M) \to \mathsf{Map}(\partial M))} \quad (23)$$

acts trivially on the boundary.

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- Therefore,

$$Z_{\mathsf{grav}}(M) = \sum_{\gamma \in \mathsf{Map}(\partial M)/\mathsf{Map}(M,\partial M)} |Z_{\mathsf{Vir}}(M^\gamma)|^2. \qquad (24)$$

Rigidity

- For hyperbolic 3-manifolds with boundaries, Map_0 is trivial.
- Therefore,

$$Z_{\mathsf{grav}}(M) = \sum_{\gamma \in \mathsf{Map}(\partial M)/\mathsf{Map}(M,\partial M)} |Z_{\mathsf{Vir}}(M^\gamma)|^2. \qquad (24)$$

- For the case without boundary, we simply get

$$Z_{\mathsf{grav}}(M) = \frac{1}{|\mathsf{Map}(M,\partial M)|} |Z_{\mathsf{Vir}}(M^\gamma)|^2. \qquad (25)$$



Figure: Cartoon by Liana Finck. May 22, 2020

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